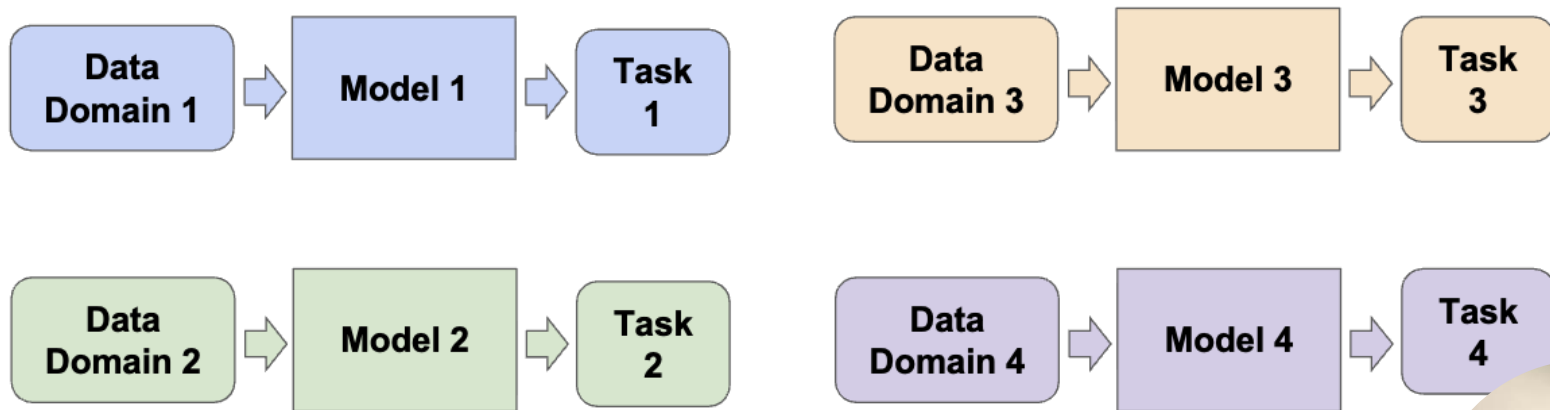


Today's Outline

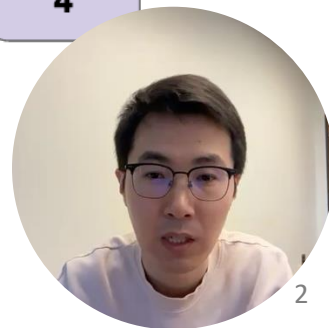
- Large Language Models and Vision-Language Models



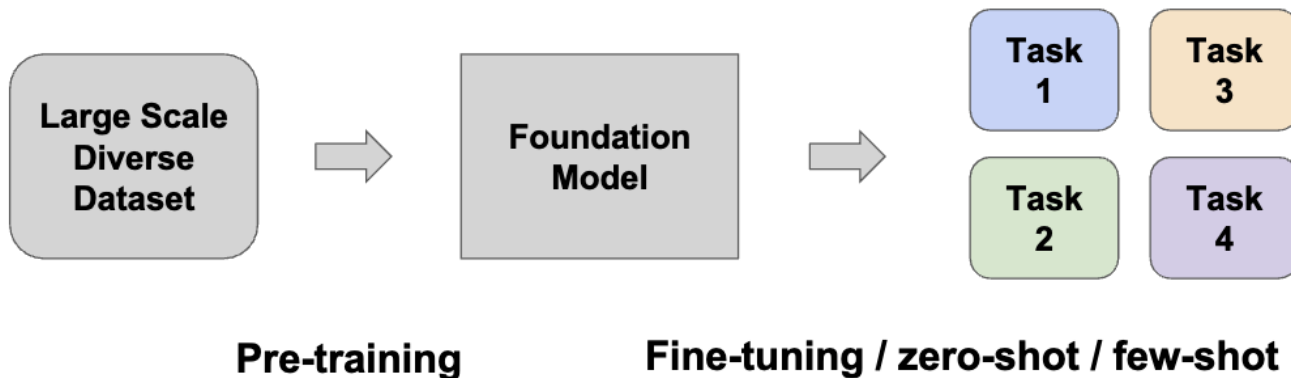
How have we been thinking about models in this class so far?



Train a specialized model for each task



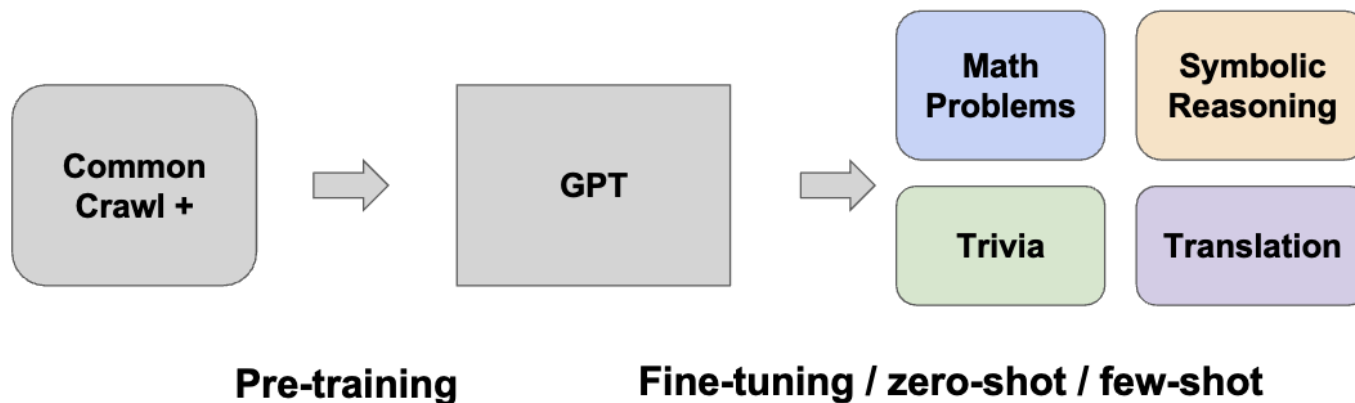
Now, we build Foundation Models



Pre-train one model that acts as the foundation for many different tasks



Foundation Models



For Language processing.



There are many classes of Foundation Models

<u>Language</u>	<u>Classification</u>	<u>LM + Vision</u>	<u>And More!</u>	<u>Chaining</u>
ELMo BERT GPT T5	CLIP CoCa	LLaVA Flamingo GPT-4V Gemini Molmo	Segment Anything Whisper Dalle Stable Diffusion Imagen	LMs + CLIP Visual Programming

Language along, and multimodal models



How do identify a model as a Foundation?

Always see with foundation models:

- general /robust to many different tasks

Often see with foundation models:

- Large # params
- Large amount of data
- Self-supervised pre-training objective



We will focus on multimodal (vision) foundation models

Language

ELMo
BERT
GPT
T5

Classification

CLIP
CoCa

LM + Vision

LLaVA
Flamingo
GPT-4V
Gemini
MoMo

And More!

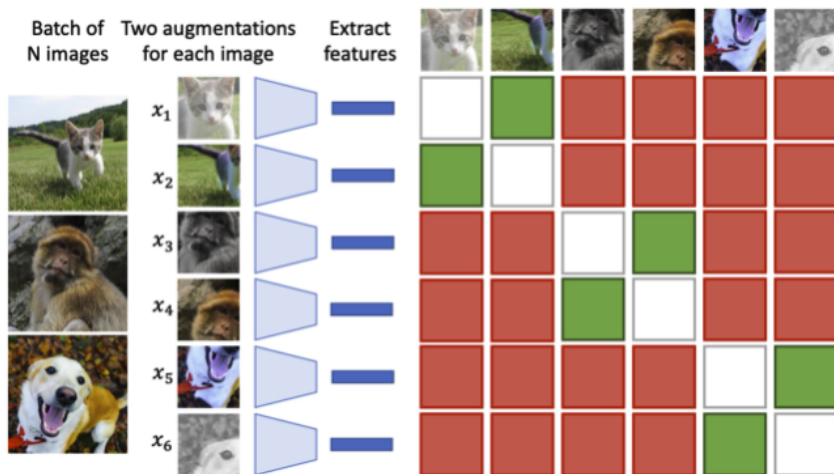
Segment Anything
Whisper
Dalle
Stable Diffusion
Imagen

Chaining

LMs + CLIP
Visual Programming



Recall this self-supervised objective from SimCLR (Contrastive Learning)

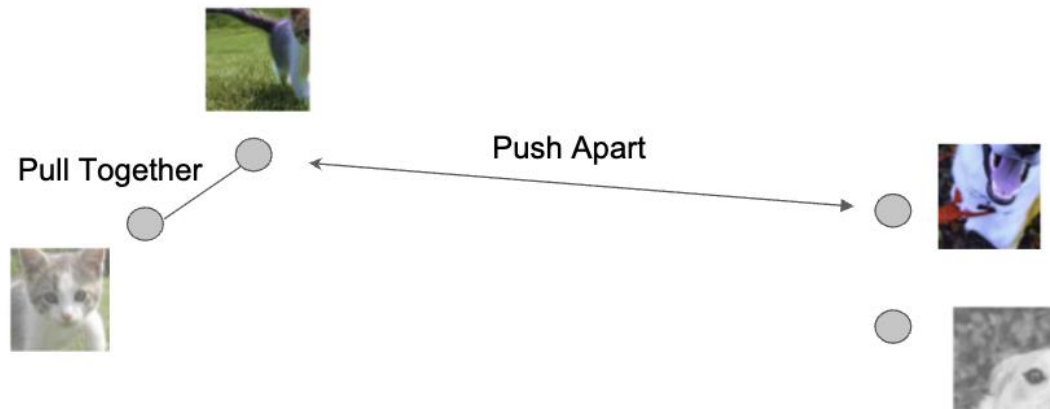


Use Self Supervised learning to learn good image features

Can train small classifiers on top of these features using supervised learning



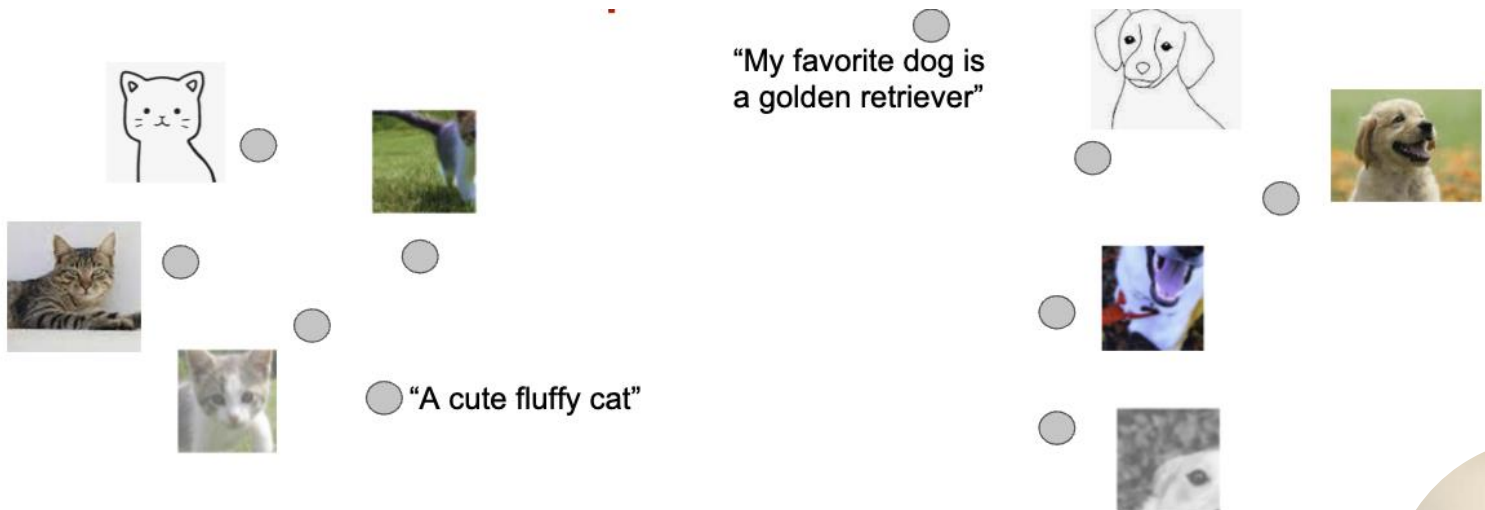
The main idea was to learning concepts without labels -> a self-supervised pretraining objective



The main idea was to learning concepts without labels -> a self-supervised pretraining objective

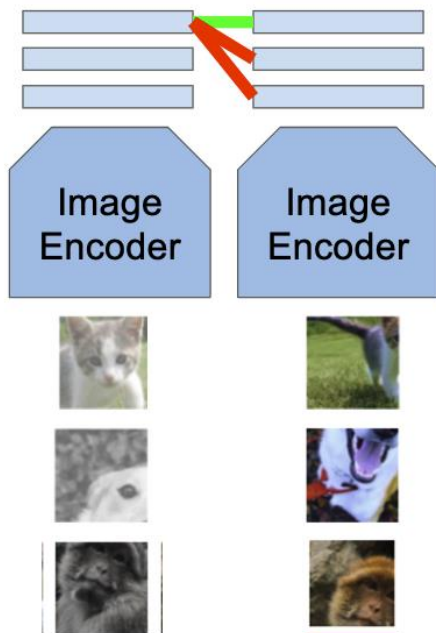


What if this representation space could also embed sentences/phrases?

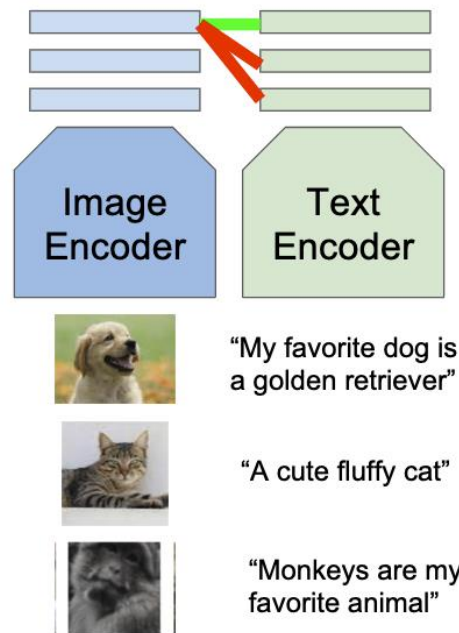


What if this representation space could also embed sentences/phrases?

SimCLR

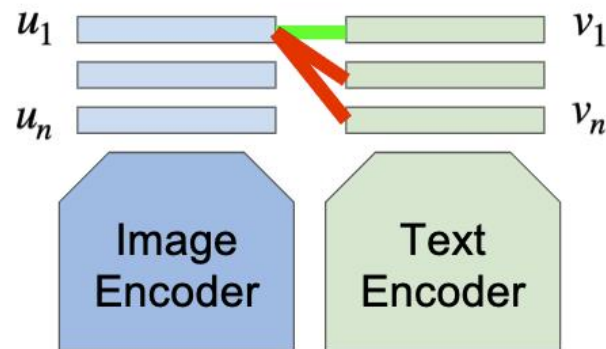


CLIP



CLIP is trained with the same contrastive objective

$$\sum_{i=1}^n -\log \left(\frac{e^{\langle u_i, v_i \rangle}}{\sum_{j=1}^n e^{\langle u_i, v_j \rangle}} \right)$$



“My favorite dog is a golden retriever”



“A cute fluffy cat”

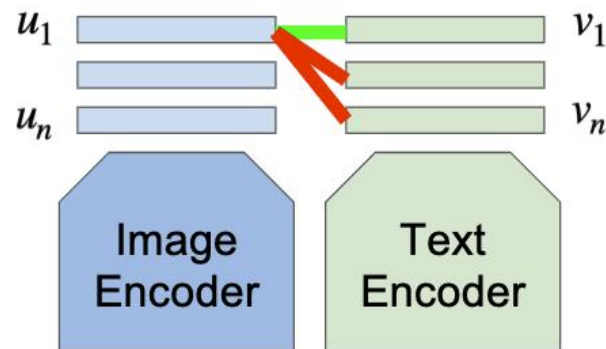


“Monkeys are favorite animal”



CLIP Training Objective

$$\sum_{i=1}^n -\log \left(\frac{e^{\langle u_i, v_i \rangle}}{\sum_{j=1}^n e^{\langle u_i, v_j \rangle}} \right) + \sum_{i=1}^n -\log \left(\frac{e^{\langle u_i, v_i \rangle}}{\sum_{j=1}^n e^{\langle u_j, v_i \rangle}} \right)$$



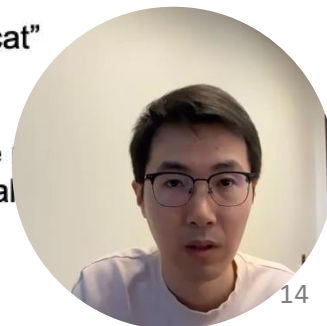
"My favorite dog is a golden retriever"



"A cute fluffy cat"



"Monkeys are favorite animal"



Lots of image-text data can be found online

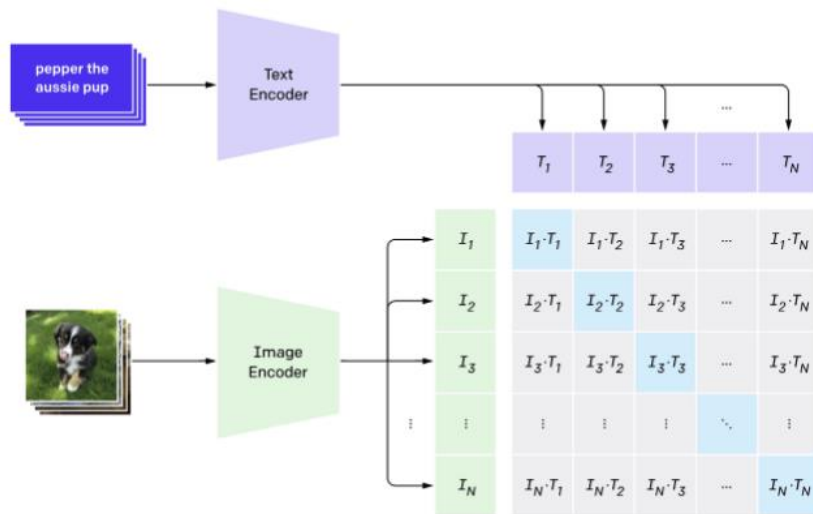


Mount Rainier's northwestern slope viewed aerially
just before sunset on September 6, 2020

CLIP training data was
scraped at scale from
images and their
associated alt-text from
the internet



CLIP Training Objective: contrastive pre-training

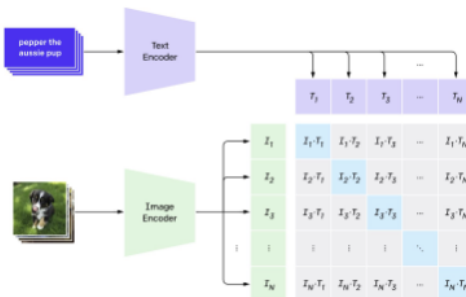


- At the end of training, you have a model that will give you a similarity score between an image and a text



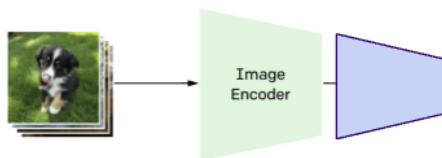
Using pre-trained models out of the box

Step 1: Pretrain a network on a pretext task that doesn't require supervision



Pre-training tasks:
Contrastive Objective

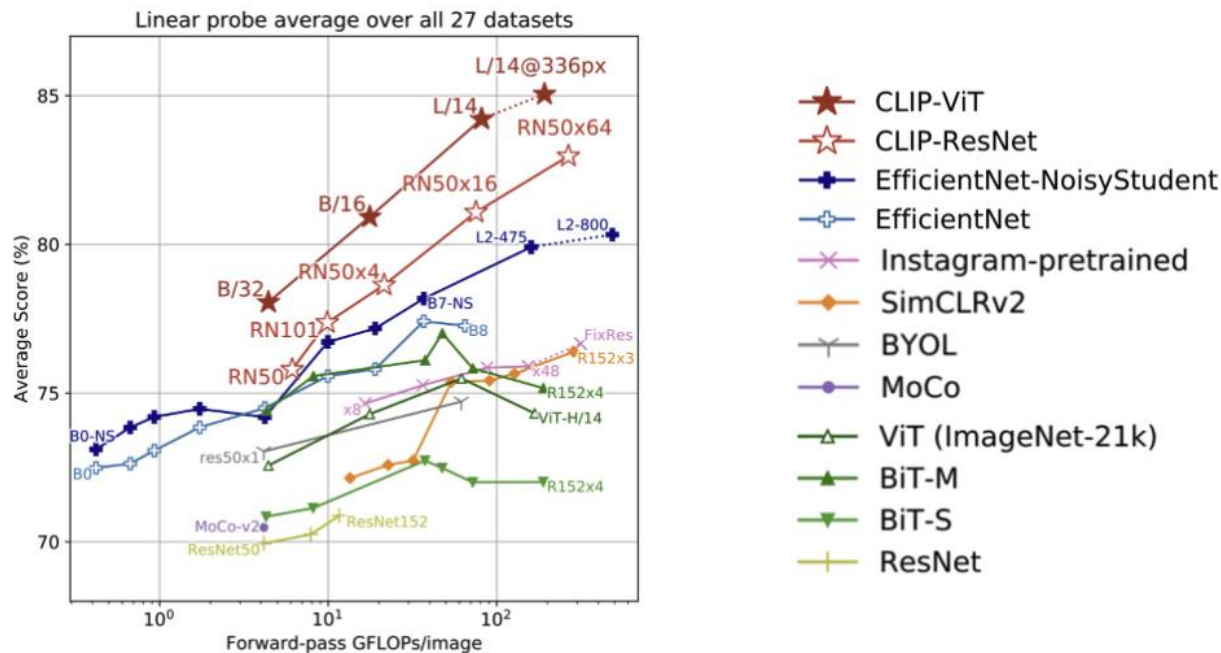
Step 2: Transfer encoder to downstream tasks via **linear classifiers**



Downstream tasks:
Image classification,
object detection,
semantic segmentation

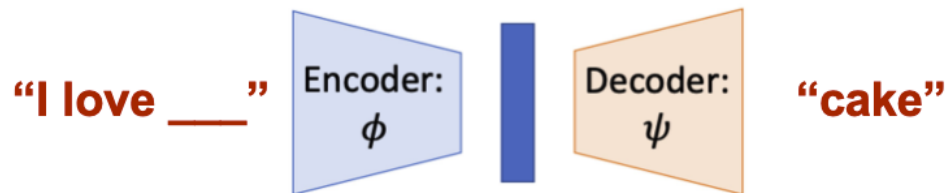


CLIP features w/ linear probe across multiple datasets

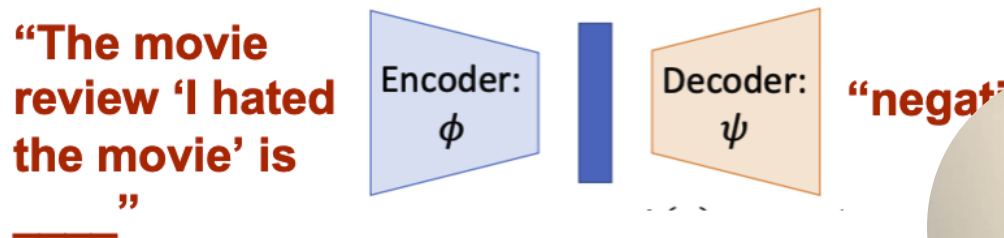


Big difference with language models: We can use LLMs zero-shot for new downstream tasks

Step 1: Pretrain a network on a pretext task that doesn't require supervision

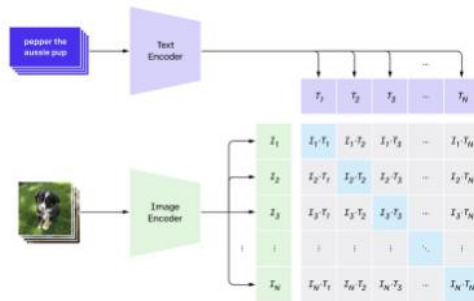


Step 2: Use the model out of the box in a creative way!



But how do we use pre-trained vision-language models in a zero-shot manner?

Step 1: Pretrain a network on a pretext task that doesn't require supervision



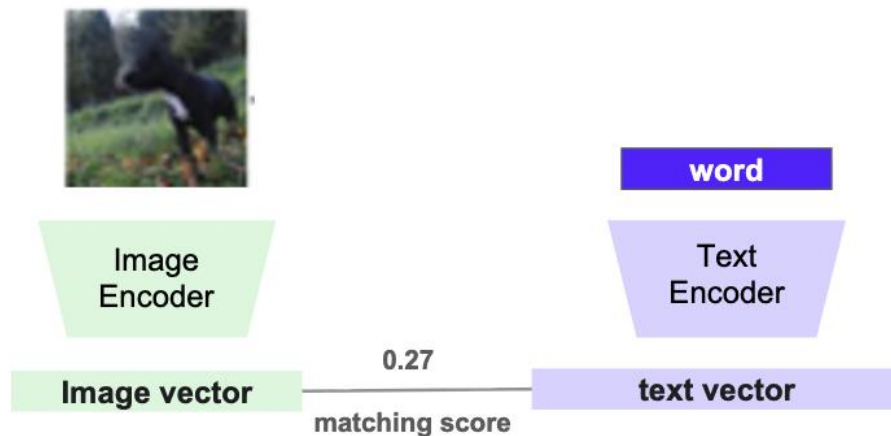
Pre-training tasks:
Contrastive Objective

Step 2: Use the model out of the box in a creative way!

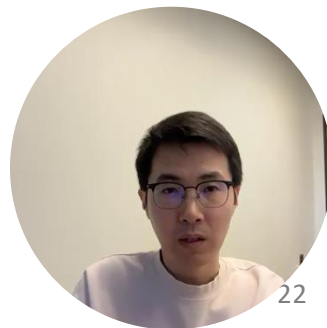
**Out of the box classification
(No fine-tuning)**



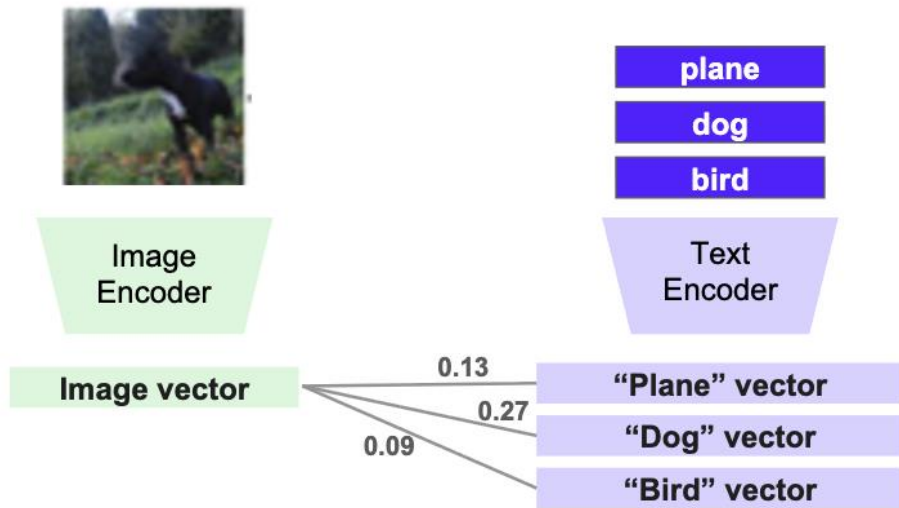
Clever trick: we can create a classifier using the text encoder!



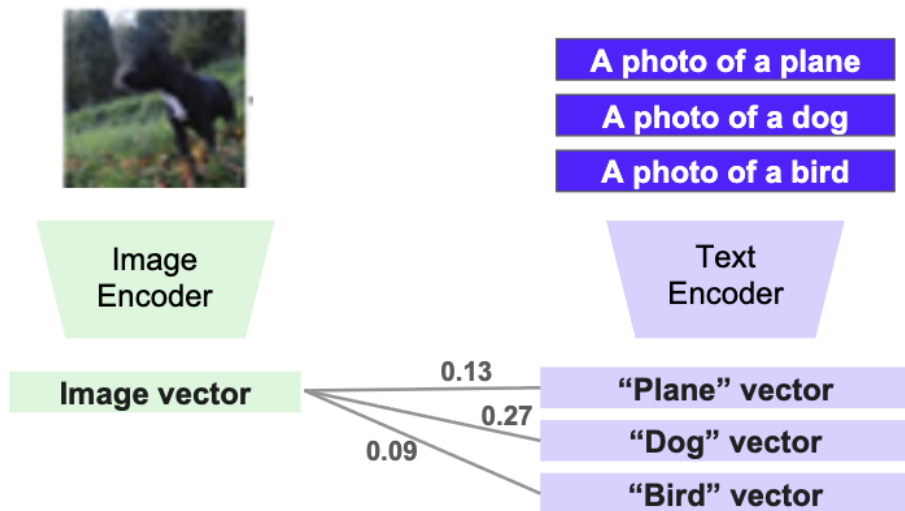
Create a vector representation for each category!



You can think of this as a 1-NN algorithm with the vectors as the training data



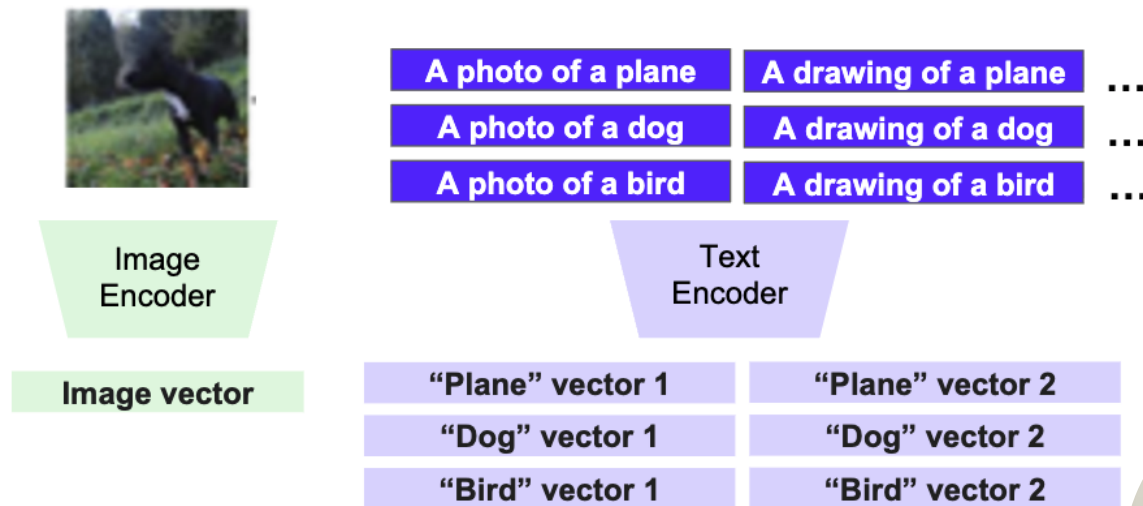
Since CLIP was trained with phrases, you can improve performance by using a phrase “A photo of a [category]”



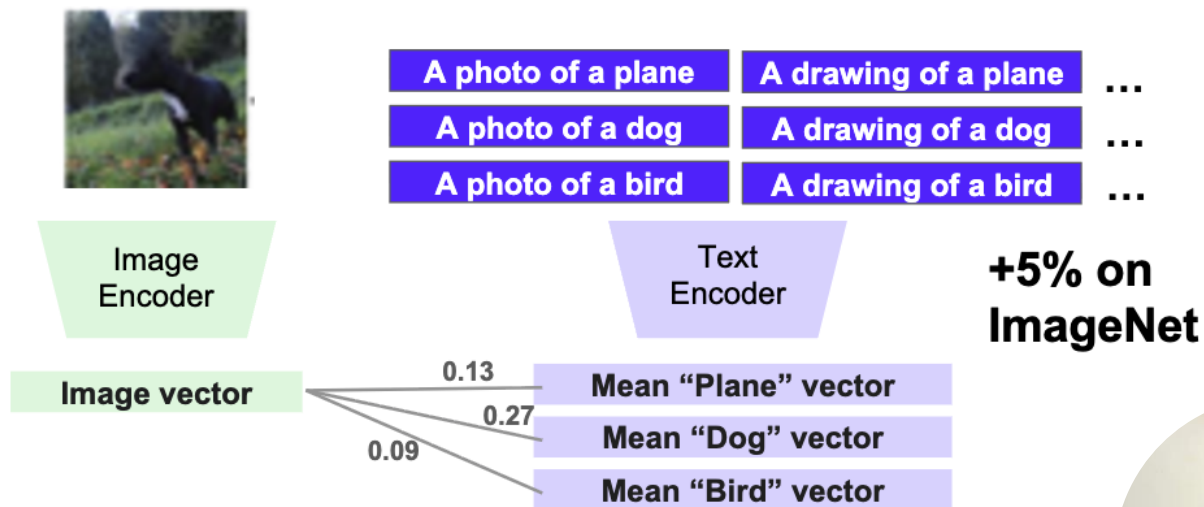
**+1.3% on
ImageNet**



A single phrase might be too biased. Solution: Use multiple phrases

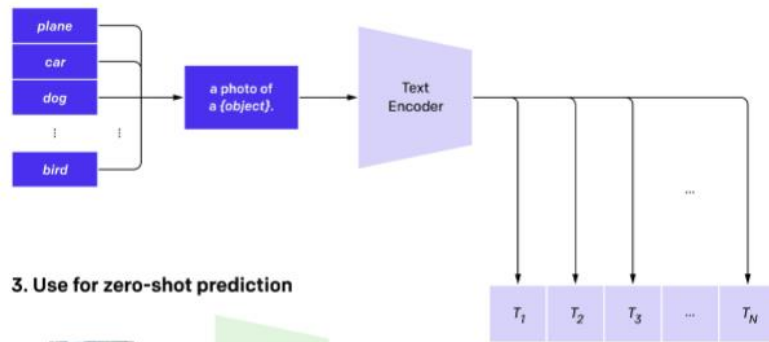


Use the average vector across phrases as the representation for each category

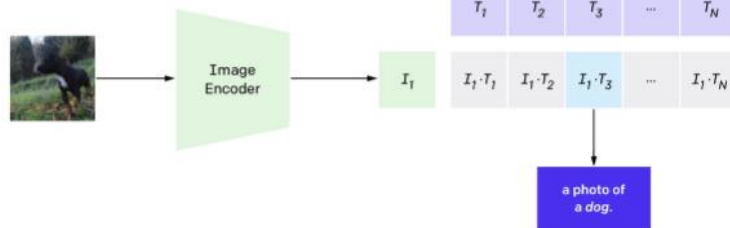


That's it! Now, you can use CLIP as a foundation model for image classification for any dataset

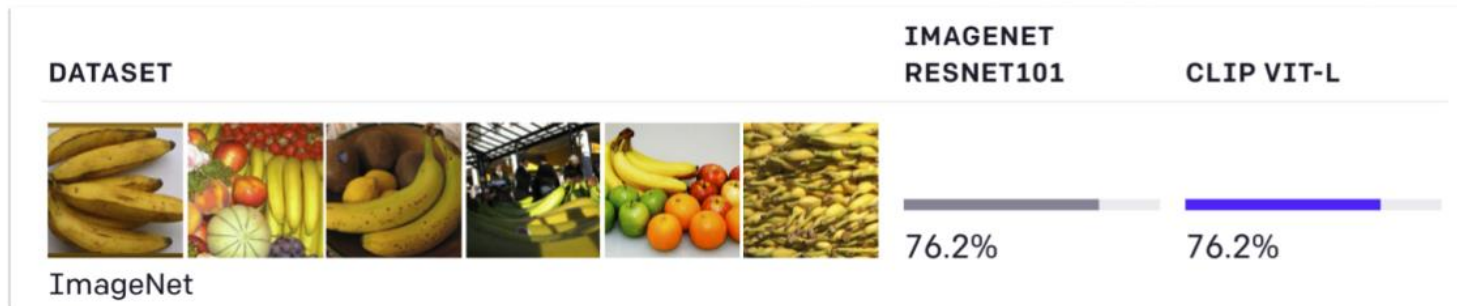
2. Create dataset classifier from label text



3. Use for zero-shot prediction



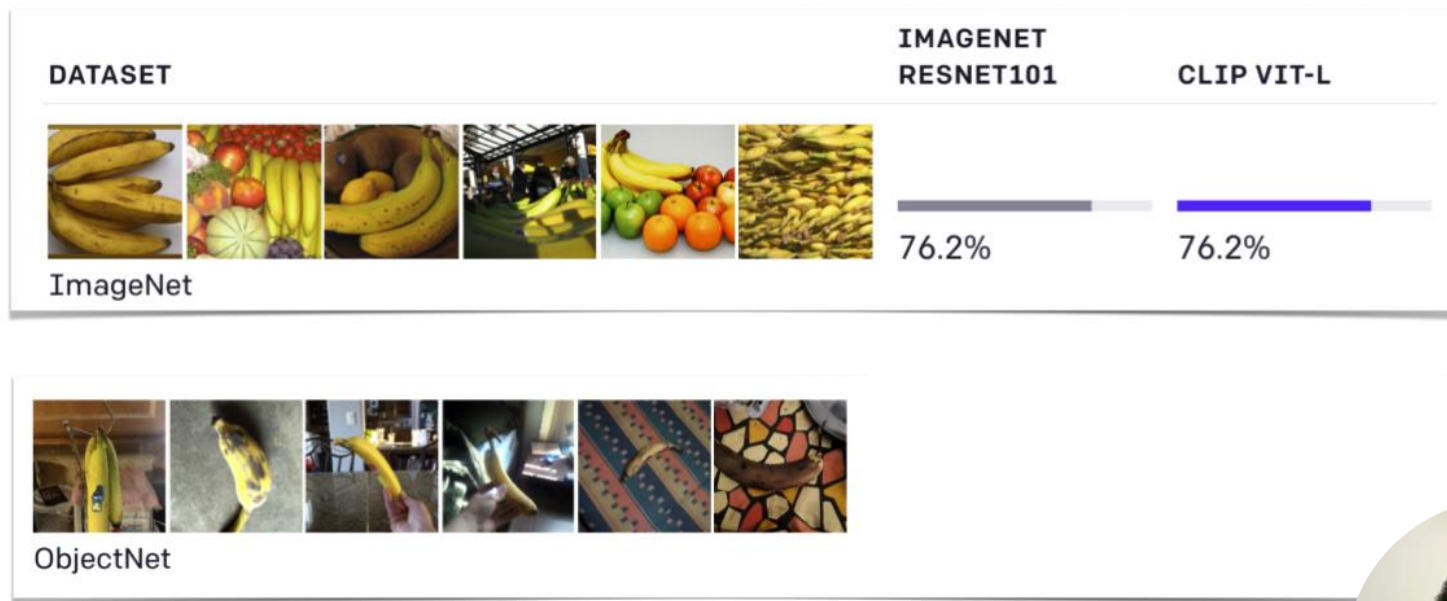
Exciting result after training on 400M image-text pairs



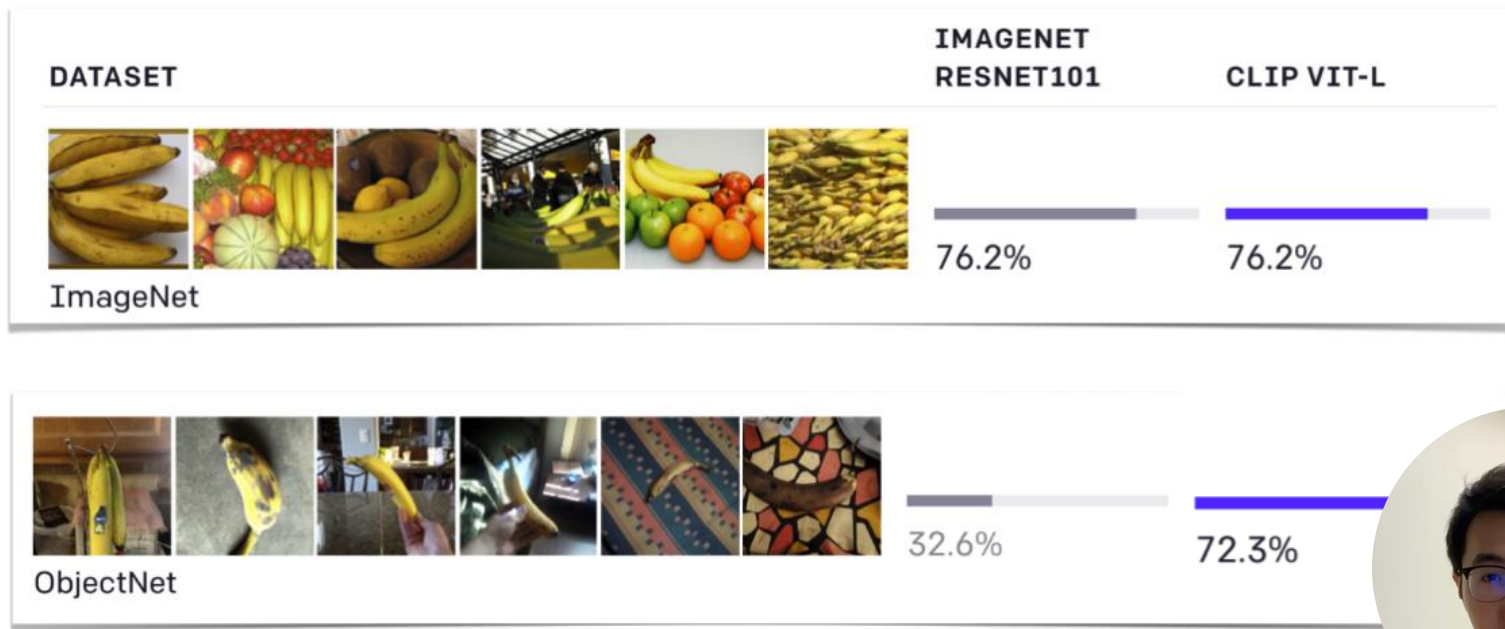
Matches the accuracy of of ResNet 101 that has been trained on ImageNet, except CLIP was trained with no human labels at all!



Here's where things get even more exciting



Training on ImageNet doesn't generalize to other datasets. ObjectNet contains the same categories but in weird viewpoints



Why does CLIP perform so well?

How can no labels beat labels??

Scale!



Foundation Models

Language

ELMo
BERT
GPT
T5

Classification

CLIP
CoCa

LM + Vision

LLaVA
Flamingo
GPT-4V
Gemini
Molmo

And More!

Segment Anything
Whisper
Dalle
Stable Diffusion
Imagen

Chaining

LMs + CLIP
Visual Programming



LLaVA

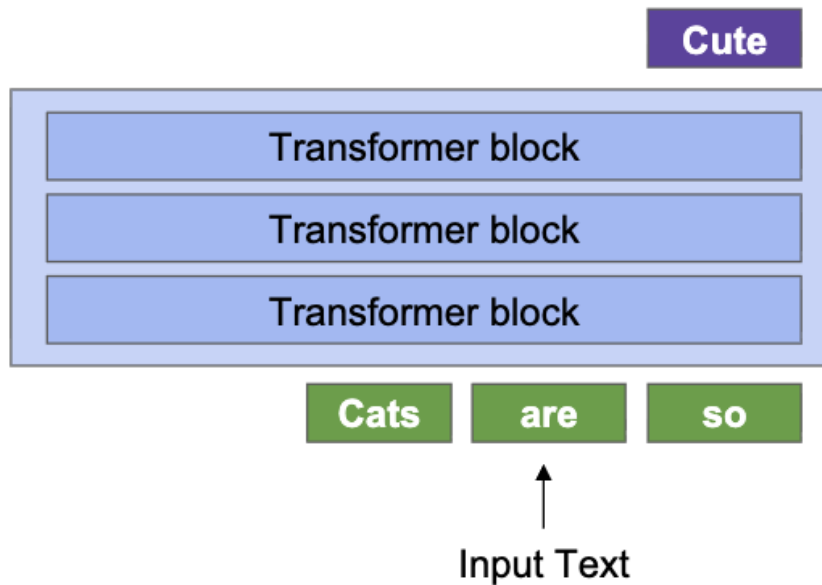
Motivation: Language models which do next token prediction can be applied to a wide variety of tasks at inference (Math, sentiment analysis, symbolic reasoning)

Can we build a model that can accept images and text as input, and then output text?

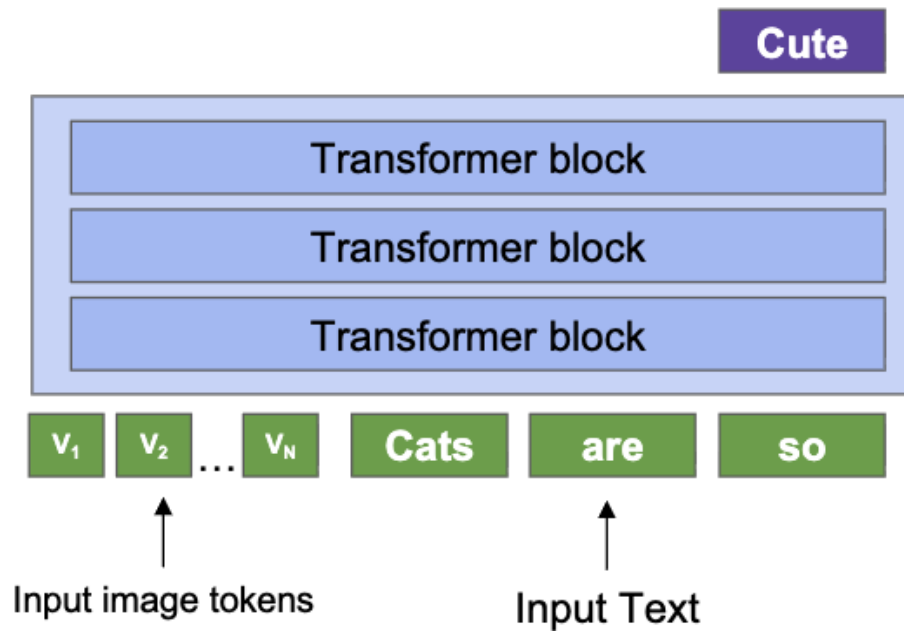
→ **Vision-Language Models**



LLaVA uses the autoregressive nature of LLMs



LLaVA uses the autoregressive nature of LLMs

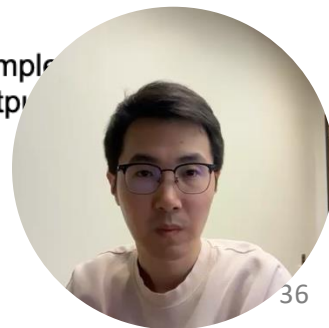
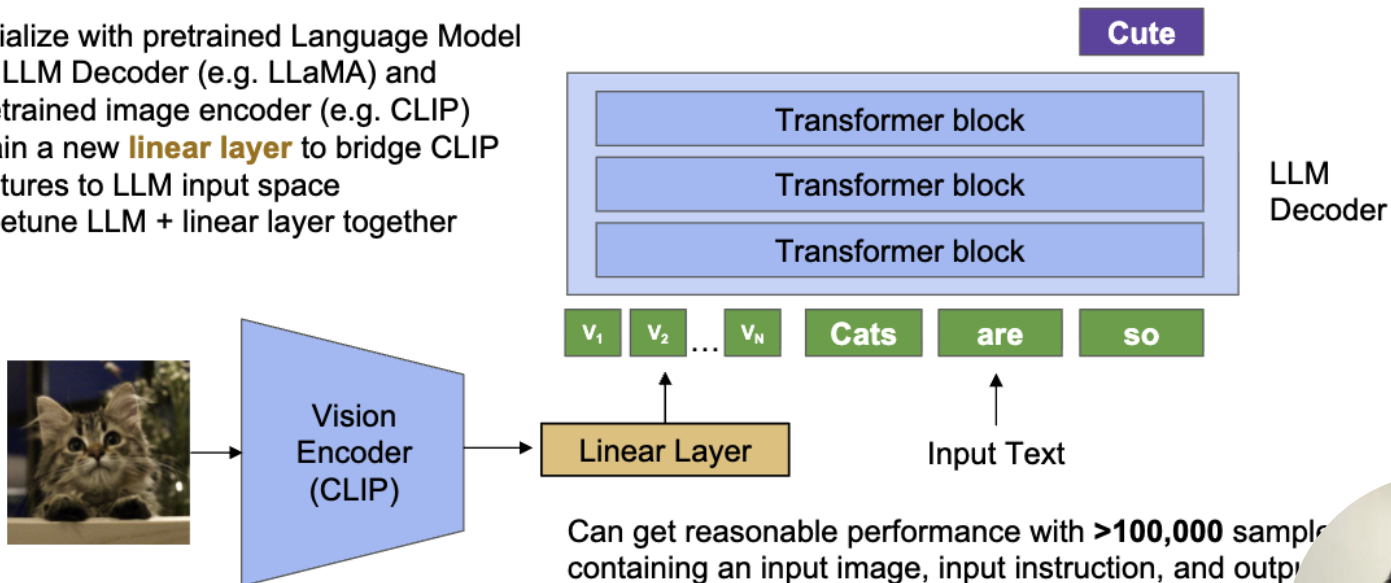


Which image tokens work best here?

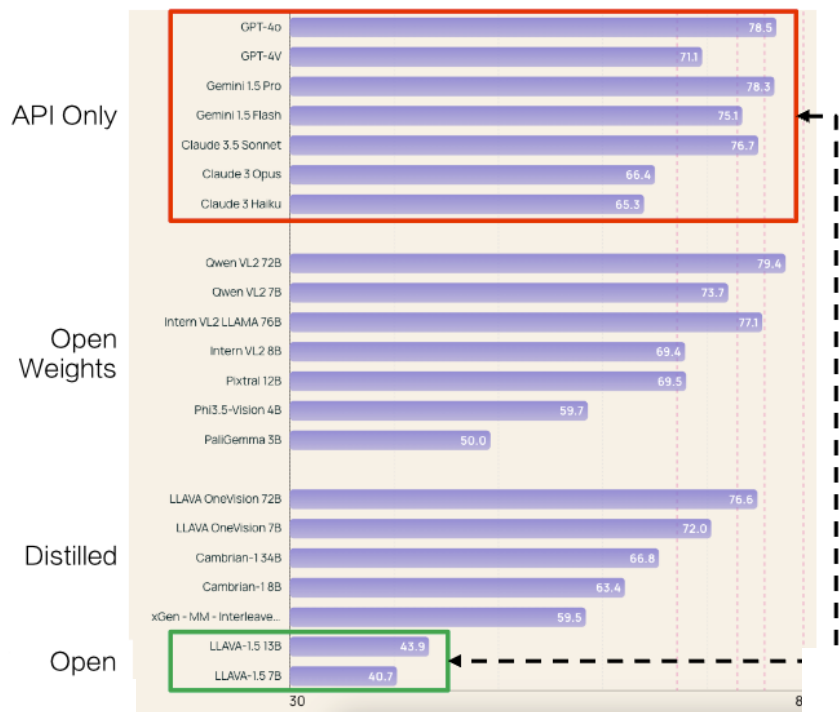


LLaVA - Overall Architecture + Training Recipe

1. Initialize with pretrained Language Model for LLM Decoder (e.g. LLaMA) and pretrained image encoder (e.g. CLIP)
2. Train a new **linear layer** to bridge CLIP features to LLM input space
3. Finetune LLM + linear layer together

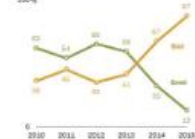


There are open-weight models but they are all distilled from GPT



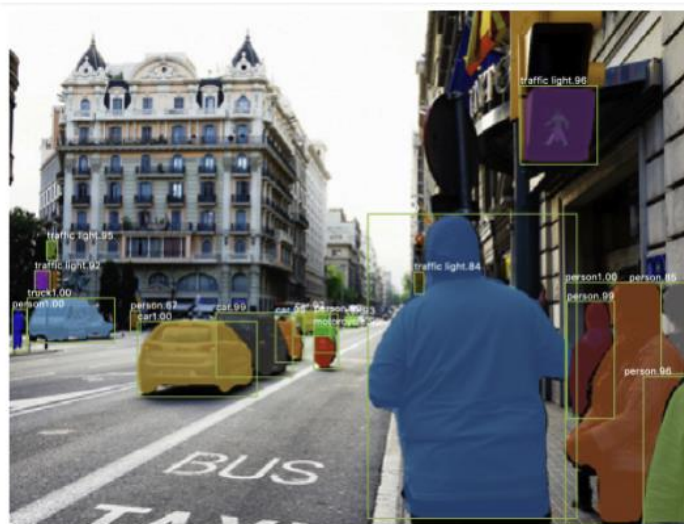
Rapid Decline in Brazilians' Assessment of Economy

Current economic situation in Brazil (%)



Segment Anything Model (SAM)

What does it mean to have a segmentation foundation model?



Masking model trained on dataset of specific number of objects (80 in COCO)

Model outputs masks of all objects in that image that is one of the categories of interest



Segment Anything Model (SAM)

What does it mean to have a segmentation foundation model?

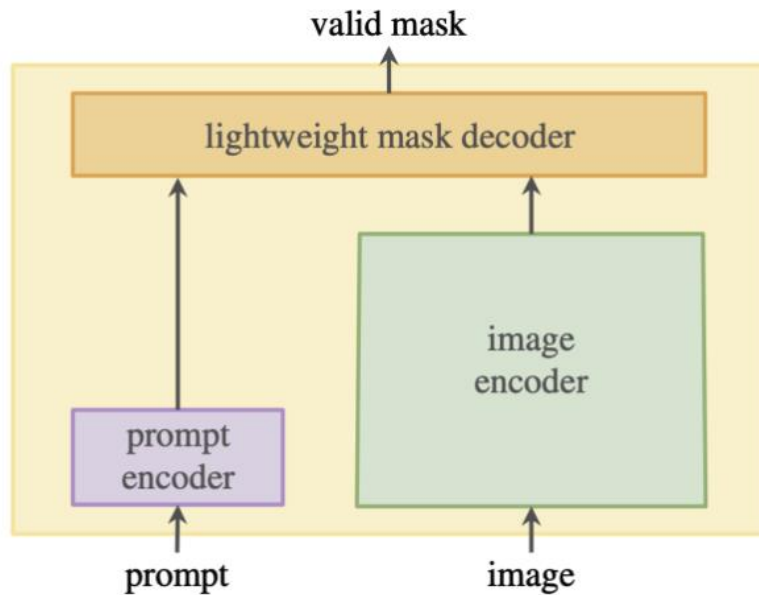


Masking model trained on a dataset of a huge number of categories

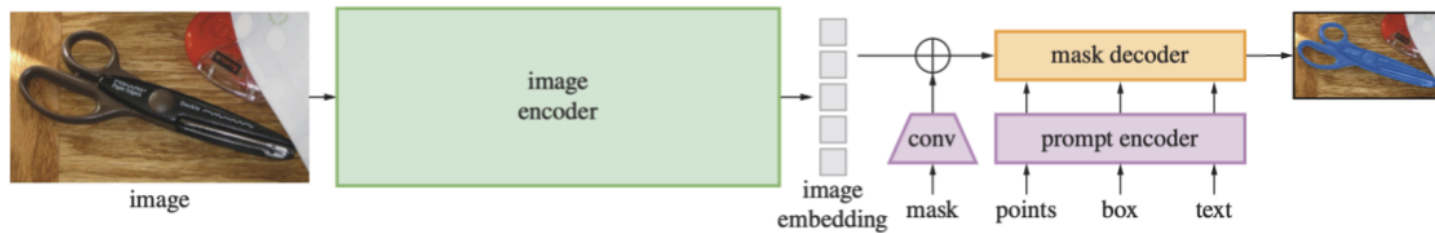
Model outputs mask of any objects that the user cares about



Basic SAM Architecture



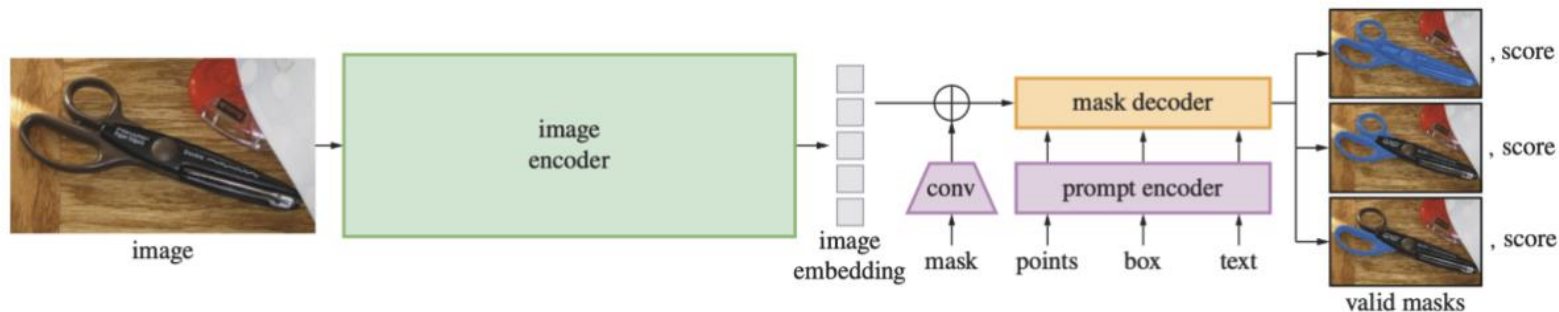
SAM Architecture



Ambiguity in correct prompt



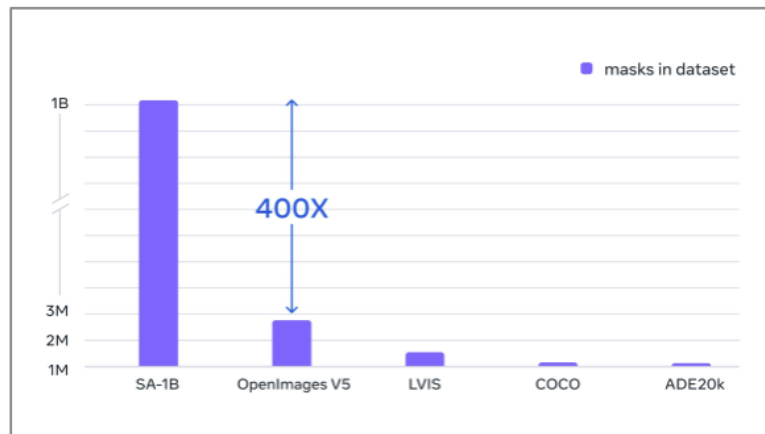
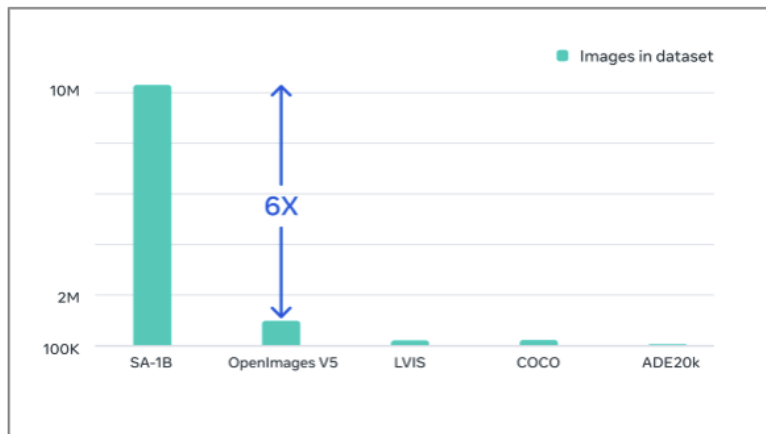
SAM Architecture



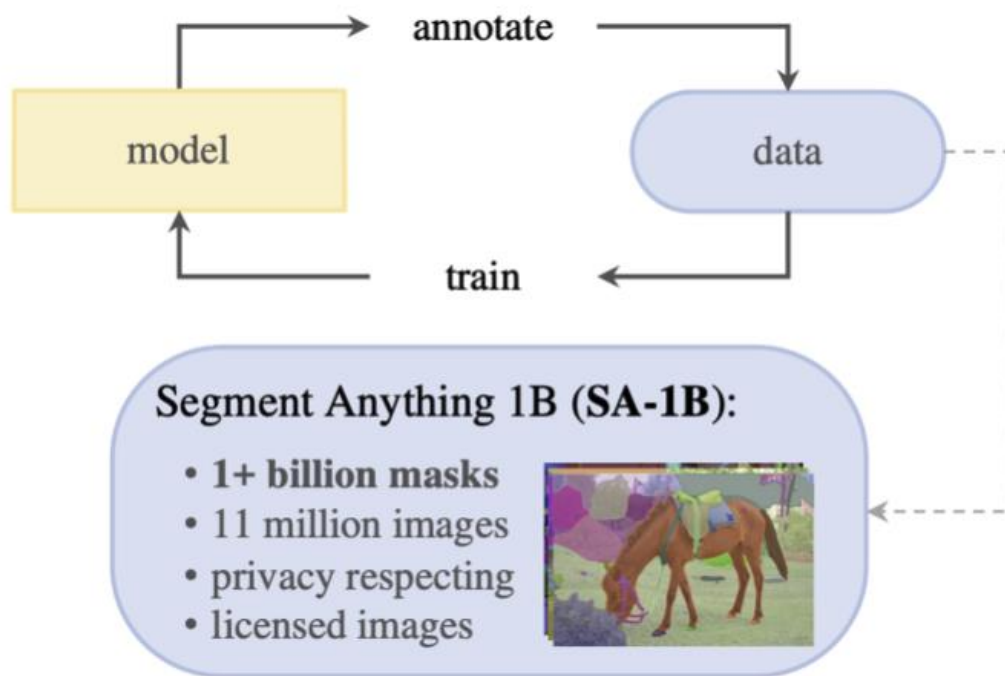
1. Loss only calculated with respect to best mask
2. Model also trained to output confidence score for each mask



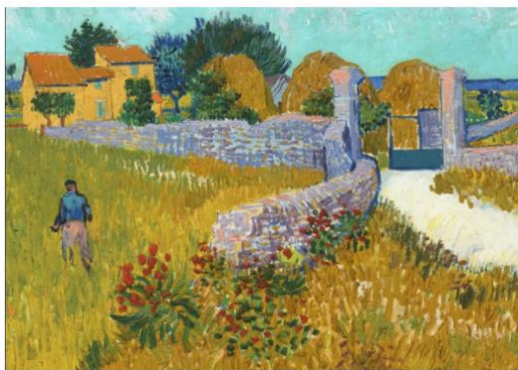
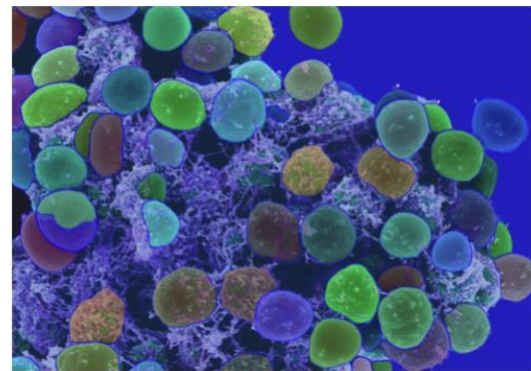
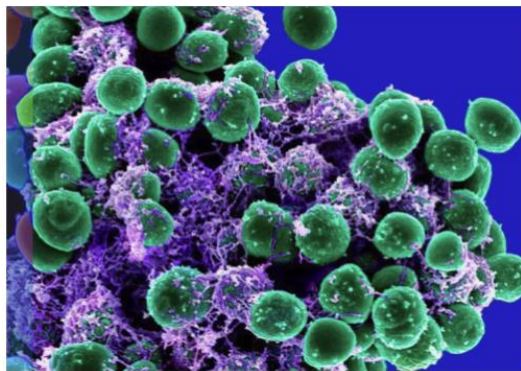
Segment Anything Model (SAM)



Segment Anything Model (SAM)



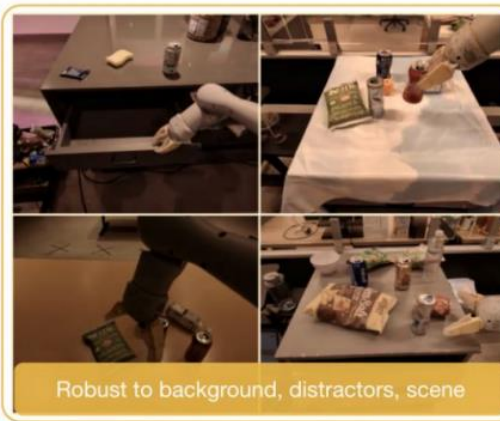
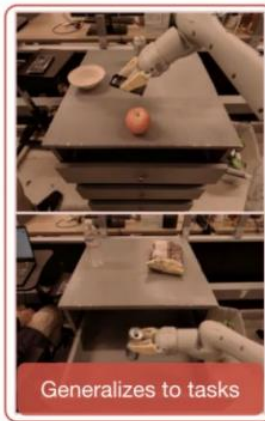
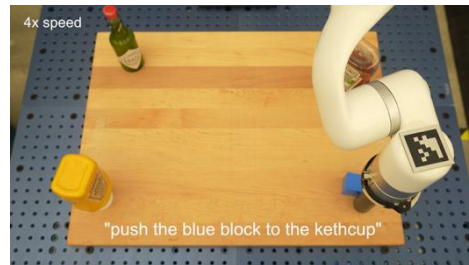
SAM Results



RT-1: Language-conditioned Imitation Learning

Method overview

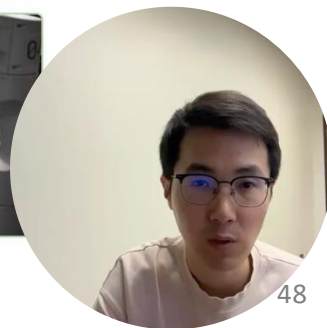
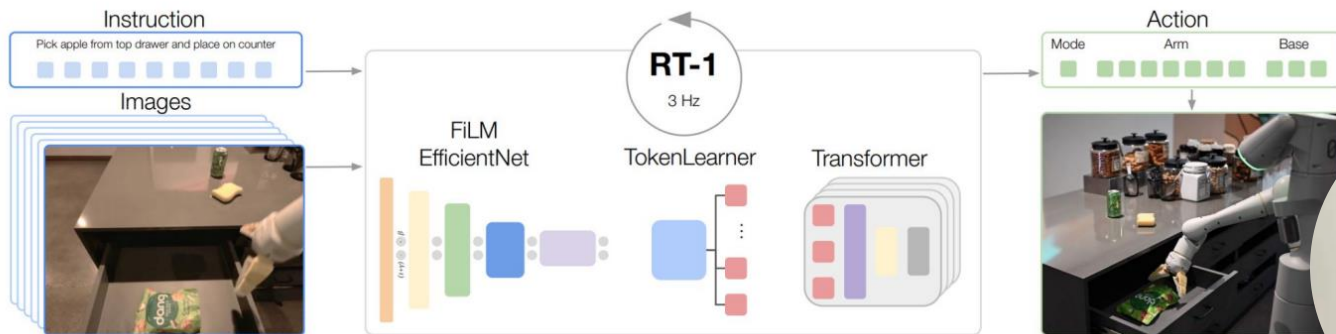
- Collected large amount of expert demonstration dataset
- Train control policy with RT-1, a transformer-based model
- Tested generalization on different tasks, changing background, etc., and long-horizon tasks
- Trained with new data sources



RT-1: Language-conditioned Imitation Learning

Model overview

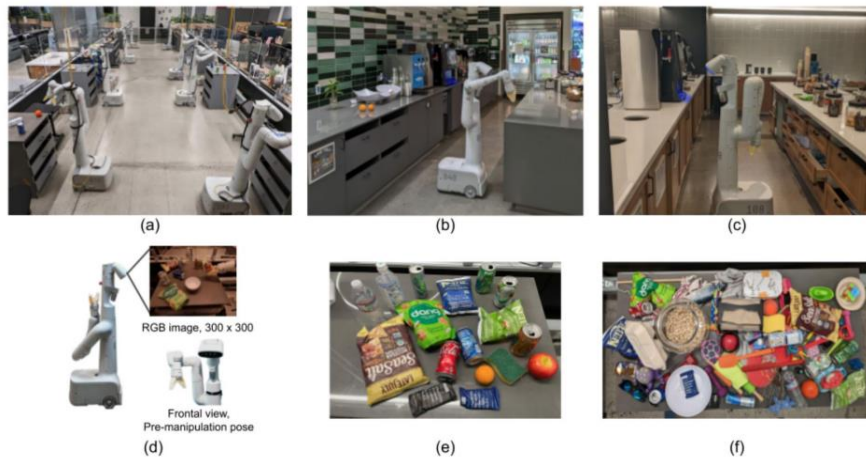
- Feed language instruction and images into RT-1
- RT-1 consists of FiLM-based EfficientNet, Token Learner and a Transformer
- FiLM: Extract visual features that are relevant to the task
- EfficientNet: ImageNet-pretrained convolutional neural network
- Token Learner: Compute a compact set of tokens
- Output 11-d action



RT-1: Language-conditioned Imitation Learning

Data overview

- 130k episodes, which contain over 700 tasks
- Collected with a fleet of 13 robots over 17 months.



RT-1: Language-conditioned Imitation Learning

Generation ability testing

- Good generalization ability in unseen tasks, changing backgrounds
- Outperforms baseline methods

Model	Seen Tasks	Unseen Tasks	Distractors	Backgrounds
Gato (Reed et al., 2022)	65	52	43	35
BC-Z (Jang et al., 2021)	72	19	47	41
BC-Z XL	56	43	23	35
RT-1 (ours)	97	76	83	59

